

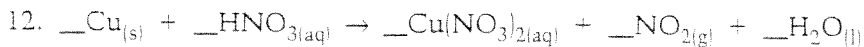
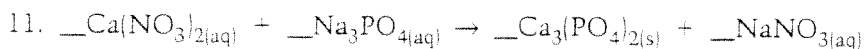
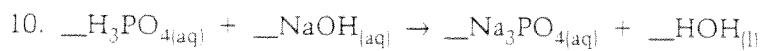
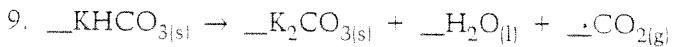
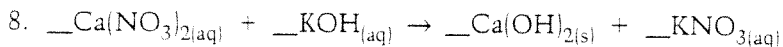
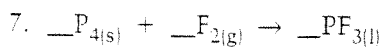
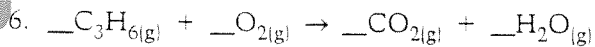
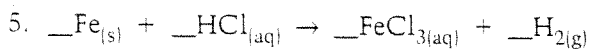
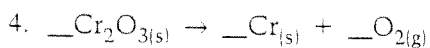
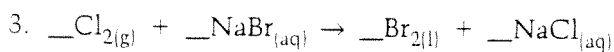
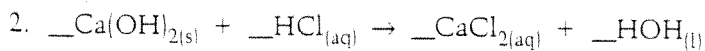
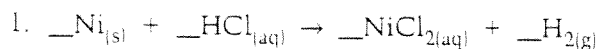
Balancing Equations #1

1. $\text{C}_3\text{H}_8 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$
2. $\text{Al}_2(\text{SO}_3)_3 + \text{NaOH} \longrightarrow \text{Na}_2\text{SO}_3 + \text{Al}(\text{OH})_3$
3. $\text{Al}_2\text{O}_3 + \text{Fe} \longrightarrow \text{Fe}_3\text{O}_4 + \text{Al}$
4. $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$
5. $\text{NH}_4\text{NO}_3 \longrightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$
6. $\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
7. $\text{P}_4\text{O}_{10} + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4$
8. $\text{Al} + \text{H}_2\text{SO}_4 \longrightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2$
9. $\text{Be}_2\text{C} + \text{H}_2\text{O} \longrightarrow \text{Be}(\text{OH})_2 + \text{CH}_4$
- 10.*** $\text{S} + \text{HNO}_3 \longrightarrow \text{H}_2\text{SO}_4 + \text{NO}_2 + \text{H}_2\text{O}$
11. $\text{NH}_3 + \text{CuO} \longrightarrow \text{Cu} + \text{N}_2 + \text{H}_2\text{O}$
- 12.*** $\text{Cu} + \text{HNO}_3 \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO} + \text{H}_2\text{O}$

Student Worksheet 5.2A

Extra Practice Questions: Balancing Chemical Equations

Balance the following equations by inspection.



Balancing Equations Race

- 1) $\text{___ C}_3\text{H}_8 + \text{___ O}_2 \rightarrow \text{___ CO}_2 + \text{___ H}_2\text{O}$
- 2) $\text{___ Al} + \text{___ Fe}_3\text{N}_2 \rightarrow \text{___ AlN} + \text{___ Fe}$
- 3) $\text{___ Na} + \text{___ Cl}_2 \rightarrow \text{___ NaCl}$
- 4) $\text{___ H}_2\text{O}_2 \rightarrow \text{___ H}_2\text{O} + \text{___ O}_2$
- 5) $\text{___ C}_6\text{H}_{12}\text{O}_6 + \text{___ O}_2 \rightarrow \text{___ H}_2\text{O} + \text{___ CO}_2$
- 6) $\text{___ H}_2\text{O} + \text{___ CO}_2 \rightarrow \text{___ C}_7\text{H}_8 + \text{___ O}_2$
- 7) $\text{___ NaClO}_3 \rightarrow \text{___ NaCl} + \text{___ O}_2$
- 8) $\text{___ (NH}_4\text{)}_3\text{PO}_4 + \text{___ Pb(NO}_3\text{)}_4 \rightarrow \text{___ Pb}_3\text{(PO}_4\text{)}_4 + \text{___ NH}_4\text{NO}_3$
- 9) $\text{___ BF}_3 + \text{___ Li}_2\text{SO}_3 \rightarrow \text{___ B}_2\text{(SO}_3\text{)}_3 + \text{___ LiF}$
- 10) $\text{___ C}_7\text{H}_{17} + \text{___ O}_2 \rightarrow \text{___ CO}_2 + \text{___ H}_2\text{O}$
- 11) $\text{___ CaCO}_3 + \text{___ H}_3\text{PO}_4 \rightarrow \text{___ Ca}_3\text{(PO}_4\text{)}_2 + \text{___ H}_2\text{CO}_3$
- 12) $\text{___ Ag}_2\text{S} \rightarrow \text{___ Ag} + \text{___ S}_8$
- 13) $\text{___ KBr} + \text{___ Fe(OH)}_3 \rightarrow \text{___ KOH} + \text{___ FeBr}_3$
- 14) $\text{___ KNO}_3 + \text{___ H}_2\text{CO}_3 \rightarrow \text{___ K}_2\text{CO}_3 + \text{___ HNO}_3$
- 15) $\text{___ Pb(OH)}_4 + \text{___ Cu}_2\text{O} \rightarrow \text{___ PbO}_2 + \text{___ CuOH}$
- 16) $\text{___ Cr(NO}_2\text{)}_2 + \text{___ (NH}_4\text{)}_2\text{SO}_4 \rightarrow \text{___ CrSO}_4 + \text{___ NH}_4\text{NO}_2$
- 17) $\text{___ KOH} + \text{___ Co}_3\text{(PO}_4\text{)}_2 \rightarrow \text{___ K}_3\text{PO}_4 + \text{___ Co(OH)}_2$
- 18) $\text{___ Sn(NO}_2\text{)}_4 + \text{___ Pt}_3\text{N}_4 \rightarrow \text{___ Sn}_3\text{N}_4 + \text{___ Pt(NO}_2\text{)}_4$
- 19) $\text{___ B}_2\text{Br}_6 + \text{___ HNO}_3 \rightarrow \text{___ B(NO}_3\text{)}_3 + \text{___ HBr}$
- 20) $\text{___ ZnS} + \text{___ AlP} \rightarrow \text{___ Zn}_3\text{P}_2 + \text{___ Al}_2\text{S}_3$

WRITE A BALANCED EQUATION FOR EACH OF THE FOLLOWING:

1. Calcium carbonate yields Calcium oxide plus carbon dioxide (CO_2)
2. Aluminum plus cupric nitrate yields aluminum nitrate plus copper.
3. Chlorine plus potassium iodide yields iodine plus potassium chloride.
4. Silver nitrate plus barium chloride yields barium nitrate plus silver chloride.
5. Mercuric oxide yields mercury plus oxygen.
6. Potassium plus water yields hydrogen plus potassium hydroxide.
7. Zinc plus sulfur (S_8) yields zinc sulfide.
8. Iron plus oxygen yields ferric oxide.
9. Chromium plus chlorine yields chromic chloride.
10. Silver plus sodium chromate yields silver chromate plus sodium.

SCH-3A Practice Writing Chemical Equations

Translate the following descriptions into a chemical equation using appropriate notations. Balance these and determine what type of reaction each one is.

1. A solid piece of iron oxidizes to form rust. Iron(II) oxide.
2. A solution of nitric acid dissolves solid copper, forming a solution of copper (II) nitrate and hydrogen gas.
3. Solutions of sulphuric acid and sodium hydroxide neutralize one another forming a solution of sodium sulphate and water.
4. When zinc is dropped into hydrochloric acid, hydrogen gas forms as well as a solution of zinc chloride.
5. When solutions of lead (II) nitrate and potassium iodide are mixed, a precipitate of lead (II) iodide and a solution of potassium nitrate forms.
6. When magnesium is heated in a strong flame, it oxidizes to magnesium oxide.
7. When an electric current is passed through molten sodium chloride, sodium metal and chlorine gas forms.
8. When Zinc and sulphur are mixed and heated in a Bunsen flame, solid zinc sulphide forms.
9. When copper(II) sulphate pentahydrate is heated, anhydrous copper (II) sulphate and water form.
10. When solutions of silver nitrate and hydrochloric acid are mixed, a white solid, silver chloride and nitric acid form.
11. When hydrogen gas and chlorine gas are mixed and treated with ultraviolet light, hydrogen chloride gas forms explosively.
12. When phosphorus is exposed to oxygen, it ignites forming diphosphorus pentaoxide.
13. When sodium metal is added to water, sodium hydroxide and hydrogen gas form.
14. When sodium carbonate is added to hydrochloric acid, carbon dioxide gas forms as well as water and a solution of sodium chloride.
15. When hydrogen and oxygen are ignited with a spark, water forms explosively.